

nb. (6.11)

Demo:	Jeune:
$L = 2m$	
$V_{min} = ?$	

То III 3.:

$$F_{max} = F_{min}$$

$$mg = m a_c \quad l.m$$

$$g = \frac{v^2}{R}$$

То же: $V^2 = Rg$

$$V_{min} = \sqrt{Rg} = \sqrt{2 \cdot 10} = \sqrt{20} \approx 4,5 \frac{m}{c}$$

Ответ: $4,5 \frac{m}{c}$

Данная задача.

№ 05.10

Jeune: 1. (2.46)

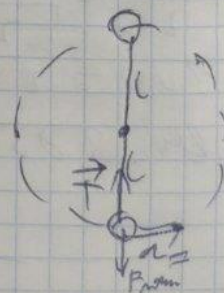
Demo:
$L = 0,5m$
$V = 10^{-1}$
$\varphi = ?$

То III 3. $T = F_{max}$

то 05: $T \cos \varphi = mg$

$$\cos \varphi = \frac{mg}{T}$$

~~$\varphi = \arccos \frac{mg}{T}$~~



$$\omega = 2\pi$$

$$\varphi = \frac{\omega}{2\pi}$$

$$\varphi = \frac{\omega}{2\pi}$$

Answer: 60

Demo:

$$V = 0,00$$

$$\varphi = \frac{\omega}{2\pi}$$

$$m = 100$$

$$F_{min} = ?$$

$$F_{max} = ?$$

$$F_{min}$$

$$\omega = 2\pi$$

$$N = m$$

$$F_{min}$$

$$= 100$$

$$F_{min}$$

Answer:

#ХАЛБАЛПАН

$$\omega = 2\pi \nu = 2\pi \frac{100}{c}$$

$$\varphi = \frac{a_y}{g} = \frac{\omega^2 R}{g} = \frac{4\pi^2 R^2}{g T^2} = \frac{4\pi^2 R^2}{g}$$

$$\varphi = \frac{4 \cdot 3,14^2 \cdot 0,5^2}{10} \approx 0,98 \text{ rad} \approx 56,6^\circ \approx 60^\circ$$

Quindi: 60°

Dato:

$$r = 0,0001 \text{ m}$$

$$V = 10 \text{ C}^{-1}$$

$$n = 100 \text{ K}$$

12. (2.48)

Domanda:



3. domanda:



$F_{min} - ?$
 $F_{max} - ?$

$$F_{grav} = mg = 1000 \text{ N} \quad T = \frac{1}{\nu} = 0,1 \text{ C}$$

$$\omega = 2\pi \nu = 20\pi \frac{1}{C}$$

$$N = ma_y; a_y = \omega^2 R$$

$$F_{grav} = m(a_y + g) = m(\omega^2 R + g) =$$

$$= 100 (400\pi^2 \cdot 0,0001 + 10) \approx 1040 \text{ N} (F_{max})$$

$$F_{min} = 100 (400\pi^2 \cdot 0,0001 - 10) \approx 960 \text{ N} (F_{min})$$

$$\text{Quindi: } 960 \text{ N e } 1040 \text{ N}$$

On 6/11

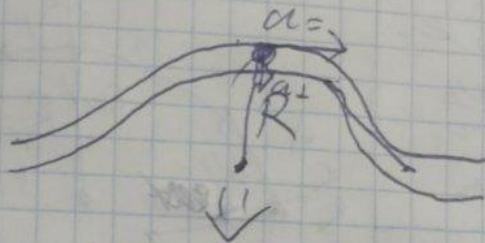
Dado:

$$R = 200 \text{ m}$$

$$\mu = 0,1$$

$$V_{\min} = ?$$

Determine:



~~$$F_{\text{mp}} = N \mu = mg \mu = 0,1 \cdot 10 \cdot m = m$$

$$a_y = \frac{v^2}{R} = \frac{v^2}{200}; F_y = m \frac{v^2}{200}$$~~

$$F_{\text{mp}} = \mu mg; F_y = \frac{mv^2}{R}$$

Como as forças se cancelam, então:

$$F_{\text{mp}} \geq F_y$$

$$\mu mg \geq \frac{mv^2}{R} \Rightarrow$$

$$\Rightarrow v \leq \sqrt{\mu g R}$$

$$v \leq \sqrt{0,1 \cdot 10 \cdot 200} = 10\sqrt{2} \approx 14,1 \frac{\text{m}}{\text{s}}$$

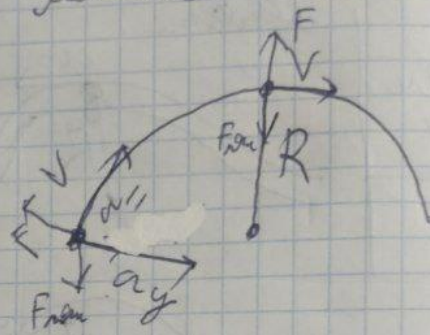
Resposta: $V_{\max} = 14,1 \frac{\text{m}}{\text{s}}$

Dado:
 $m = 2500 \text{ kg}$
 $V = 400 \frac{\text{km}}{\text{h}}$
 $R = 500 \text{ m}$

$\varphi = ?$ $F = ?$

15. (2.59)

Terminale:



$$F_{\text{grav}} = 25000 \text{ N} = 25 \text{ kN}$$

$$a_y = \frac{V^2}{R} = \frac{400^2}{500} = 320 \frac{\text{km}^2}{\text{h}^2}$$

$$O_x: F \sin \varphi = m a_y$$

$$O_y: F \cos \varphi = m g \approx 0$$

$$\frac{O_x}{O_y} = \tan \varphi = \frac{a_y}{g} = \frac{V^2}{R g}$$

$$\tan^2 \varphi = \frac{1}{\cos^2 \varphi} + 1 \Rightarrow$$

$$\Rightarrow \frac{1}{\cos^2 \varphi} = \tan^2 \varphi + 1$$

$$\cos \varphi = \sqrt{\frac{1}{\tan^2 \varphi + 1}}$$

$$\text{Když } F = \frac{m g}{\cos \varphi} = \frac{m g}{R g} \sqrt{R^2 g^2 + V^4} =$$

$$= \frac{m}{R} R g V^2 = m g V^2 =$$

$$= 2500 \cdot 10 \cdot 400^2 = 4000000000 = 4 \text{ TH}$$

$$\varphi = \arctg \frac{400^2}{500 \cdot 10} = \arctg \frac{160000}{5000} =$$

$$\approx \arctg 32 \approx 88,2^\circ$$

$$\text{Given: } \varphi = 88,2^\circ, F = 4 \text{ TH}$$

Distance $\frac{1}{10}$ of λ

on $10, 10$

$$\lambda \approx 7,43$$

Given:

$$L = 0,5 \text{ m}$$

$$m = 0,36 \text{ kg}$$

$$d = \frac{1}{12} \text{ m}$$

$J = ?$

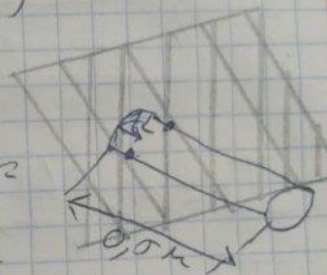
Then:

$$J_{cm} = \frac{1}{12} m L^2 =$$

$$= \frac{0,36 \cdot 0,25}{12} =$$

$$= 0,0025 \frac{\text{kg} \cdot \text{m}^2}{12}$$

$\approx 0,0025 \frac{\text{kg} \cdot \text{m}^2}{12}$



$$J = J_{cm} + m d^2; J = 0,0025 + 0,36 \cdot \frac{1}{144} \approx$$

$$\approx 0,01 \frac{\text{kg} \cdot \text{m}^2}{12}$$

$$\text{Answer: } J \approx 0,0070 \frac{\text{kg} \cdot \text{m}^2}{12}$$